

~\Downloads\Day 16.sql

```
1  CREATE OR REPLACE DATABASE COMPANY_SECURITY;
2
3  USE DATABASE COMPANY_SECURITY;
4
5  CREATE OR REPLACE SCHEMA HR;
6
7  USE SCHEMA HR;
8
9  CREATE OR REPLACE TABLE EMPLOYEES
10 (
11     EMP_ID INT,
12     EMP_NAME STRING,
13     DEPARTMENT STRING,
14     SALARY NUMBER,
15     EMAIL STRING
16 );
17
18
19 INSERT INTO EMPLOYEES VALUES
20 (101, 'John', 'HR', 70000, 'john@company.com'),
21 (102, 'Alice', 'HR', 65000, 'alice@company.com'),
22 (103, 'David', 'IT', 80000, 'david@company.com'),
23 (104, 'Neha', 'IT', 75000, 'neha@company.com'),
24 (105, 'Ravi', 'Finance', 90000, 'ravi@company.com'),
25 (106, 'Priya', 'Finance', 85000, 'priya@company.com'),
26 (107, 'Akash', 'Sales', 60000, 'akash@company.com'),
27 (108, 'Sneha', 'Sales', 62000, 'sneha@company.com');
28
29 CREATE ROLE HR_MANAGER;
30
31 CREATE ROLE HR_ROLE;
32
33 CREATE ROLE IT_ROLE;
34
35 CREATE ROLE FINANCE_ROLE;
36
37 CREATE ROLE SALES_ROLE;
38
39 GRANT USAGE ON DATABASE COMPANY_SECURITY TO ROLE HR_MANAGER;
40 GRANT USAGE ON DATABASE COMPANY_SECURITY TO ROLE HR_ROLE;
41
42 GRANT USAGE ON SCHEMA HR TO ROLE HR_MANAGER;
43 GRANT USAGE ON SCHEMA HR TO ROLE HR_ROLE;
44
45 GRANT SELECT ON TABLE EMPLOYEES TO ROLE HR_MANAGER;
46 GRANT SELECT ON TABLE EMPLOYEES TO ROLE HR_ROLE;
47
48 CREATE OR REPLACE MASKING POLICY SALARY_MASK
```

```
49 AS (SALARY NUMBER)
50 RETURNS NUMBER ->
51 CASE
52     WHEN CURRENT_ROLE()='HR_MANAGER'
53     THEN SALARY
54     ELSE NULL
55 END;
56
57 ALTER TABLE EMPLOYEES
58 MODIFY COLUMN SALARY
59 SET MASKING POLICY SALARY_MASK;
60
61 CREATE OR REPLACE ROW ACCESS POLICY DEPT_POLICY
62 AS (DEPARTMENT STRING)
63 RETURNS BOOLEAN ->
64 CASE
65
66 WHEN CURRENT_ROLE()='HR_MANAGER'
67 THEN TRUE
68
69 WHEN CURRENT_ROLE()='HR_ROLE'
70 THEN DEPARTMENT='HR'
71
72 WHEN CURRENT_ROLE()='IT_ROLE'
73 THEN DEPARTMENT='IT'
74
75 WHEN CURRENT_ROLE()='FINANCE_ROLE'
76 THEN DEPARTMENT='Finance'
77
78 WHEN CURRENT_ROLE()='SALES_ROLE'
79 THEN DEPARTMENT='Sales'
80
81 ELSE FALSE
82
83 END;
84
85 ALTER TABLE EMPLOYEES
86 ADD ROW ACCESS POLICY DEPT_POLICY
87 ON (DEPARTMENT);
88
89 USE ROLE HR_MANAGER;
90
91 SELECT *
92 FROM EMPLOYEES;
93
94 USE ROLE IT_ROLE;
95
96 SELECT *
97 FROM EMPLOYEES;
98
```

```
99  -- Task 2
100 CREATE OR REPLACE DATABASE SALES_DB;
101
102 USE DATABASE SALES_DB;
103
104 CREATE TABLE CUSTOMERS
105 (
106  CUSTOMER_ID INT,
107  CUSTOMER_NAME STRING,
108  CITY STRING,
109  COUNTRY STRING
110 );
111
112 INSERT INTO CUSTOMERS VALUES
113 (1,'Rahul','Mumbai','India'),
114 (2,'Priya','Pune','India'),
115 (3,'John','New York','USA'),
116 (4,'David','Chicago','USA'),
117 (5,'Ravi','Delhi','India'),
118 (6,'Neha','Jaipur','India'),
119 (7,'Alice','London','UK'),
120 (8,'Tom','Manchester','UK'),
121 (9,'Akash','Nagpur','India'),
122 (10,'Sneha','Hyderabad','India');
123
124 CREATE ROLE ANALYST_ROLE;
125
126 CREATE ROLE DEVELOPER_ROLE;
127
128 CREATE USER ANALYST1
129  PASSWORD='Password@123'
130  DEFAULT_ROLE=ANALYST_ROLE
131  MUST_CHANGE_PASSWORD=TRUE;
132
133 CREATE USER DEV1
134  PASSWORD='Password@123'
135  MUST_CHANGE_PASSWORD=TRUE;
136
137 GRANT SELECT
138  ON TABLE CUSTOMERS
139  TO ROLE ANALYST_ROLE;
140
141 GRANT ROLE ANALYST_ROLE
142  TO USER ANALYST1;
143
144 -- UBAC
145 GRANT SELECT
146  ON TABLE CUSTOMERS
147  TO USER DEV1;
148
```

```
149 REVOKE SELECT
150 ON TABLE CUSTOMERS
151 FROM ROLE ANALYST_ROLE;
152
153 SELECT *
154 FROM CUSTOMERS;
```